

SDS 8102: Advanced Mathematical Statistics

Homework 2 — Solutions

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1. Unbiasedness, Completeness, and Admissibility in the Gaussian Model

1. A complete sufficient admissible statistic.

The joint density is

$$f_\mu(x_1, \dots, x_n) = (2\pi)^{-n/2} \exp \left\{ -\frac{1}{2} \sum_{i=1}^n x_i^2 + \mu \sum_{i=1}^n x_i - \frac{n\mu^2}{2} \right\}.$$

By the factorization theorem, $\sum_i X_i$, and equivalently $T = \bar{X}$, is sufficient for μ .

This is a full-rank one-parameter exponential family: the natural parameter space is all of $\mathbb{R}$, which contains an open interval. The standard completeness theorem for full-rank exponential families therefore implies that T is complete. Equivalently, if $\mathbb{E}_\mu[g(T)] = 0$ for every μ (with the required integrability), the uniqueness of the Laplace transform applied to

$$0 = e^{-n\mu^2/2} \mathbb{E}_0[g(T)e^{n\mu T}]$$

gives $g(T) = 0$ almost surely.

The word *admissible* is understood here for $T = \bar{X}$ viewed as an estimator of μ under squared-error loss. Put $v = 1/n$. For the proper prior $\mu \sim N(0, \tau^2)$, the Bayes estimator is

$$\delta_\tau(T) = a_\tau T, \quad a_\tau = \frac{\tau^2}{\tau^2 + v}.$$

Since a posterior mean minimizes posterior squared error,

$$r(\pi_\tau, T) - r(\pi_\tau, \delta_\tau) = \mathbb{E}_{\pi_\tau}[(T - \delta_\tau(T))^2] = \frac{v^2}{\tau^2 + v} = O(\tau^{-2}). \quad (1)$$

If an estimator δ dominated T , with a strict improvement at some μ_0 , the usual continuity argument for Gaussian risks would give an interval I containing μ_0 and an $\varepsilon > 0$ such that

$$R(\mu, T) - R(\mu, \delta) \geq \varepsilon, \quad \mu \in I.$$

Consequently,

$$r(\pi_\tau, T) - r(\pi_\tau, \delta) \geq \varepsilon \pi_\tau(I) = \Omega(\tau^{-1}). \quad (2)$$

But Bayes optimality of δ_τ implies that the left side of (2) is at most (1), which is $O(\tau^{-2})$. This is impossible for large τ . Hence $\bar{X}$ is admissible.

2. Analyticity of Gaussian expectations.

For $X \sim N(\mu, 1)$, change of measure from P_0 gives

$$m(\mu) = e^{-\mu^2/2} \mathbb{E}_0[g(X)e^{\mu X}]. \quad (3)$$

For any $r > 0$,

$$e^{r|X|} \leq e^{rX} + e^{-rX}.$$

Moreover,

$$\mathbb{E}_0[|g(X)|e^{\pm rX}] = e^{r^2/2} \mathbb{E}_{\pm r}[|g(X)|] < \infty.$$

Thus $|g(X)|e^{r|X|}$ is integrable. If $|\mu| \leq r$, then

$$|g(X)| \sum_{k=0}^{\infty} \frac{|\mu X|^k}{k!} = |g(X)|e^{|\mu X|} \leq |g(X)|e^{r|X|}.$$

Dominated convergence therefore permits termwise expectation in (3):

$$m(\mu) = e^{-\mu^2/2} \sum_{k=0}^{\infty} \frac{\mu^k}{k!} \mathbb{E}_0[g(X)X^k]. \quad (4)$$

Indeed, for every $r > 0$,

$$\sum_{k=0}^{\infty} \frac{r^k}{k!} \left| \mathbb{E}_0[g(X)X^k] \right| \leq \mathbb{E}_0[|g(X)|e^{r|X|}] < \infty.$$

Hence the power series in (4) has infinite radius of convergence. Its product with $e^{-\mu^2/2}$ is real analytic, so m is differentiable on $\mathbb{R}$. If $g(X)$ were unbiased for $|\mu|$, then $m(\mu) = |\mu|$ for every μ , contradicting nondifferentiability of $|\mu|$ at zero. Therefore there is no integrable unbiased estimator of $|\mu|$.

3. Admissibility without unbiasedness (Bonus).

For $\delta_0(X) = 0$,

$$R(\mu, \delta_0) = \mu^2, \quad R(0, \delta_0) = 0.$$

If δ dominated δ_0 , then

$$0 \leq R(0, \delta) = \mathbb{E}_0[\delta(X)^2] \leq R(0, \delta_0) = 0.$$

Thus $\delta(X) = 0$ P_0 -almost surely. Since all Gaussian location distributions have strictly positive densities on the same sample space, P_μ and P_0 are mutually absolutely continuous. Hence $\delta(X) = 0$ P_μ -almost surely for every μ . Any purported dominator is therefore equivalent to δ_0 and cannot improve strictly anywhere. Thus δ_0 is admissible. This also shows that admissibility does not imply unbiasedness (or even good risk away from a particular parameter value).

2. Two-Sided Testing and Confidence Intervals in the Gaussian Model

1. The likelihood-ratio test.

Apart from constants, the log-likelihood is

$$\ell(\mu) = -\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2,$$

and its unrestricted maximizer is $\hat{\mu} = \bar{X}$. Therefore

$$-2 \log \Lambda = \frac{1}{\sigma^2} \left\{ \sum_{i=1}^n (X_i - \mu_0)^2 - \sum_{i=1}^n (X_i - \bar{X})^2 \right\} = \frac{n(\bar{X} - \mu_0)^2}{\sigma^2}.$$

The LRT rejects for large values of this statistic, equivalently when

$$\frac{\sqrt{n}|\bar{X} - \mu_0|}{\sigma} > z_{1-\alpha/2}.$$

Under H_0 ,

$$Z = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} \sim N(0, 1),$$

so its rejection probability is $\mathbb{P}(|Z| > z_{1-\alpha/2}) = \alpha$.

2. Inverting the test.

A value μ_0 is accepted precisely when

$$\frac{\sqrt{n}|\bar{X} - \mu_0|}{\sigma} \leq z_{1-\alpha/2}.$$

Solving this inequality for μ_0 gives

$$C(X) = \left[\bar{X} - z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}} \right].$$

For the true value μ ,

$$\mathbb{P}_\mu \{ \mu \in C(X) \} = \mathbb{P}_\mu \left(\frac{\sqrt{n}|\bar{X} - \mu|}{\sigma} \leq z_{1-\alpha/2} \right) = 1 - \alpha.$$

3. Why the test is not UMP.

For a fixed $\mu_1 > \mu_0$, the likelihood ratio is

$$\frac{f_{\mu_1}(x)}{f_{\mu_0}(x)} = C \exp \left\{ \frac{n(\mu_1 - \mu_0)}{\sigma^2} \bar{x} \right\},$$

which is strictly increasing in $\bar{x}$. The Neyman–Pearson test is therefore, uniquely up to null sets,

$$\mathbf{1} \left\{ \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} > z_{1-\alpha} \right\}.$$

For $\mu_1 < \mu_0$, the ratio is strictly decreasing and the unique most powerful test rejects when

$$\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} < -z_{1-\alpha}.$$

A UMP test for the two-sided alternative would have to coincide almost surely with both distinct tests. This is impossible. Hence no UMP level- α test exists for the two-sided alternative.

3. Bayes Estimation and the Impossibility of an Unbiased Bayes Rule

1. The Bayes estimator.

Completing the square gives

$$\theta \mid X = x \sim N\left(\frac{\tau^2}{1 + \tau^2}x, \frac{\tau^2}{1 + \tau^2}\right).$$

The Bayes estimator under squared-error loss is the posterior mean,

$$\delta_\tau(X) = \frac{\tau^2}{1 + \tau^2}X.$$

The posterior variance is constant, so the Bayes risk is

$$r(\pi, \delta_\tau) = \mathbb{E}[\text{Var}(\Theta \mid X)] = \frac{\tau^2}{1 + \tau^2}.$$

2. Frequentist bias.

For fixed θ ,

$$\mathbb{E}_\theta[\delta_\tau(X)] = \frac{\tau^2}{1 + \tau^2}\theta, \quad \text{Bias}_\theta(\delta_\tau) = -\frac{\theta}{1 + \tau^2}.$$

Thus it is unbiased only at $\theta = 0$. As $\tau^2 \rightarrow \infty$, $\delta_\tau(X) \rightarrow X$, which is unbiased. However, the limiting flat prior $\pi(\theta) \propto 1$ is improper, so this limit is only generalized Bayes, not Bayes under a proper prior.

3. Can a proper Bayes estimator also be unbiased?

Write $\delta_\pi(X) = \mathbb{E}[\Theta \mid X]$. Conditional expectation gives

$$\mathbb{E}[(\delta_\pi(X) - \Theta)\delta_\pi(X)] = \mathbb{E}[\delta_\pi(X)\{\delta_\pi(X) - \mathbb{E}[\Theta \mid X]\}] = 0. \quad (5)$$

On the other hand, assumed frequentist unbiasedness gives

$$\mathbb{E}[(\delta_\pi(X) - \Theta)\Theta] = \int \theta\{\mathbb{E}_\theta[\delta_\pi(X)] - \theta\} d\pi(\theta) = 0. \quad (6)$$

Subtracting (6) from (5),

$$\mathbb{E}[(\delta_\pi(X) - \Theta)^2] = 0.$$

Consequently $\delta_\pi(X) = \Theta$ almost surely. In the Gaussian model, however,

$$\text{Var}(\Theta \mid X) = \frac{\tau^2}{1 + \tau^2} > 0,$$

so X does not determine Θ exactly. This is a contradiction. More generally, an unbiased estimator can also be proper Bayes under squared error only in the degenerate situation in which the posterior identifies the parameter with probability one.

4. Bayesian Interpretations of Ridge Regression and the Lasso

1. Ridge regression from a Gaussian prior.

The likelihood and prior yield

$$-\log \pi(\beta \mid Y) = \frac{1}{2\sigma^2} \|Y - X\beta\|_2^2 + \frac{1}{2\tau^2} \|\beta\|_2^2 + \text{constant}.$$

Multiplying by σ^2 shows that the posterior mode minimizes

$$\frac{1}{2} \|Y - Xb\|_2^2 + \frac{\lambda}{2} \|b\|_2^2, \quad \lambda = \frac{\sigma^2}{\tau^2}.$$

The normal equations give

$$\hat{\beta}_{\text{ridge}} = (X^\top X + \lambda I_p)^{-1} X^\top Y.$$

Completing the square also gives

$$\beta \mid Y \sim N_p \left((X^\top X + \lambda I_p)^{-1} X^\top Y, \sigma^2 (X^\top X + \lambda I_p)^{-1} \right).$$

A Gaussian density is symmetric about its mean, which is also its mode; hence the posterior mean and posterior mode both equal the ridge estimator.

2. The lasso from a Laplace prior.

Independence and the Laplace density imply

$$-\log \pi(\beta \mid Y) = \frac{1}{2\sigma^2} \|Y - X\beta\|_2^2 + \frac{1}{b} \|\beta\|_1 + \text{constant}.$$

After multiplication by σ^2 , the posterior mode is

$$\hat{\beta}_{\text{lasso}} = \arg \min_{u \in \mathbb{R}^p} \left\{ \frac{1}{2} \|Y - Xu\|_2^2 + \lambda \|u\|_1 \right\}, \quad \lambda = \frac{\sigma^2}{b}.$$

3. Shrinkage under an orthogonal design.

When $X^\top X = I_p$ and $Z = X^\top Y$,

$$\hat{\beta}_{\text{ridge}} = \frac{Z}{1 + \lambda}.$$

For the lasso, discard terms independent of u to write the objective as

$$\sum_{j=1}^p \left\{ \frac{1}{2} (u_j - Z_j)^2 + \lambda |u_j| \right\}.$$

If $u_j > 0$, the stationarity equation gives $u_j = Z_j - \lambda$; if $u_j < 0$, it gives $u_j = Z_j + \lambda$; and $u_j = 0$ is optimal when $|Z_j| \leq \lambda$. Therefore

$$\hat{\beta}_{\text{lasso},j} = \text{sign}(Z_j)(|Z_j| - \lambda)_+.$$

Ridge multiplies every coordinate by the same nonzero shrinkage factor and therefore does not ordinarily produce exact zeros. Lasso uses soft thresholding and sets every coordinate with $|Z_j| \leq \lambda$ exactly to zero.

4. Posterior mean versus posterior mode (Bonus).

Under squared-error loss, the posterior expected loss satisfies

$$\mathbb{E}[\|\beta - a\|_2^2 \mid Y] = \mathbb{E}[\|\beta - \mathbb{E}[\beta \mid Y]\|_2^2 \mid Y] + \|a - \mathbb{E}[\beta \mid Y]\|_2^2.$$

It is minimized by the posterior mean. Thus ridge is a Bayes estimator under squared-error loss because its Gaussian posterior mean equals its mode. With a Laplace prior, the posterior is generally non-Gaussian and asymmetric, so its mean generally differs from its mode. The lasso is the posterior mode (MAP), but it is not generally the Bayes estimator under squared-error loss.

Finally, a continuous posterior may have its density maximized at $\beta_j = 0$ even though the singleton $\{\beta_j = 0\}$ has posterior probability zero. A mode concerns the height of a density, not the probability mass of a single point, so there is no contradiction.